UPSKILL SPRINT · WALL REFERENCE · LEAN SIX SIGMA & QUALITY

The Excel Statistics Line Map

Every statistical function in Excel as a station. Board at your question, change at an ⇄ interchange, arrive at the answer.

111 STATISTICAL · **21** HELPERS · **27** TOOLS · **39** LEGACY · **12** GAPS

12 LINES — 01 Descriptive · 02 Distributions · 03 Inference · 04 Hypothesis tests · 05 Correlation · 06 Regression · 07 Forecasting · 08 EDA · 09 What-if · 10 ToolPak · 11 Legacy · 12 Beyond Excel

○ Interchange

01 DESCRIPTIVE STATISTICS

Describe what I have · 36 stations

- **AVERAGE**(number1, [number2], ...) — Default centre of a data set
- **AVERAGEA**(value1, [value2], ...) — Only when logical values must count
- **AVERAGEIF**(range, criteria, [average_range]) — Mean for one shift, grade or line
- **AVERAGEIFS**(average_range, criteria_range1, criteria1, ...) — Mean by heat and orientation together
- **MEDIAN**(number1, [number2], ...) — Skewed data such as delays and cycle times
- **MODE.SNGL**(number1, [number2], ...) — Discrete readings with repeats
- **MODE.MULT**(number1, [number2], ...) — Data with two or more peaks
- **GEOMEAN**(number1, [number2], ...) — Growth rates, ratios, log-normal data
- **HARMEAN**(number1, [number2], ...) — Averaging rates such as units per hour
- **TRIMMEAN**(array, percent) — A robust mean when outliers are present
- **STDEV.S**(number1, [number2], ...) — Almost every quality calculation on sample data
- **STDEV.P**(number1, [number2], ...) — Only when you hold the whole population
- **STDEVA**(value1, [value2], ...) — Rarely; text becomes 0 and distorts results
- **STDEVPA**(value1, [value2], ...) — Rarely; same caution as STDEVA
- **VAR.S**(number1, [number2], ...) — ANOVA, F-tests and pooled variance
- **VAR.P**(number1, [number2], ...) — Complete populations only
- **VARA**(value1, [value2], ...) — Rarely
- **VARPA**(value1, [value2], ...) — Rarely
- **AVEDEV**(number1, [number2], ...) — A spread summary that resists outliers
- **DEVSQ**(number1, [number2], ...) — Building ANOVA sums of squares by hand
- **SKEW**(number1, [number2], ...) — Quick normality and symmetry screen
- **SKEW.P**(number1, [number2], ...) — Skewness of a complete population
- **KURT**(number1, [number2], ...) — Screening for heavy or light tails
- **MAX**(number1, [number2], ...) — Range and worst-case checks
- **MAXA**(value1, [value2], ...) — Rarely
- **MAXIFS**(max_range, criteria_range1, criteria1, ...) — Worst reading per grade or line
- **MIN**(number1, [number2], ...) — Range and minimum-spec checks
- **MINA**(value1, [value2], ...) — Rarely
- **MINIFS**(min_range, criteria_range1, criteria1, ...) — Lowest toughness per heat
- **LARGE**(array, k) — Top-k values and order statistics
- **SMALL**(array, k) — Bottom-k values and order statistics
- **COUNT**(value1, [value2], ...) — The sample size for every other formula
- **COUNTA**(value1, [value2], ...) — Checking for text entries in numeric columns
- **COUNTBLANK**(range) — Missing-data audit
- **COUNTIF**(range, criteria) — Defects beyond a spec limit
- **COUNTIFS**(criteria_range1, criteria1, ...) — Contingency tables and DPU by category

02 PROBABILITY DISTRIBUTIONS

How likely is it? · 41 stations

- **NORM.DIST**(x, mean, standard_dev, cumulative) — Share of output below a limit, ppm out of spec
- ○ **NORM.INV**(probability, mean, standard_dev) ⇄ 09 — Percentile limits; simulation with RAND()
- **NORM.S.DIST**(z, cumulative) — p-values for z statistics
- ○ **NORM.S.INV**(probability) ⇄ 03 — Critical z, sample size, sigma level
- **GAUSS**(z) — Quick area from the centre
- **PHI**(x) — Plotting the bell curve
- **T.DIST**(x, deg_freedom, cumulative) — Lower-tailed t tests
- **T.DIST.RT**(x, deg_freedom) — Upper-tailed t tests
- ○ **T.DIST.2T**(x, deg_freedom) ⇄ 04 — Two-sided p-value for a t statistic
- **T.INV**(probability, deg_freedom) — One-sided critical values; negative for small p
- ○ **T.INV.2T**(probability, deg_freedom) ⇄ 04 — Confidence intervals and two-sided tests
- **CHISQ.DIST**(x, deg_freedom, cumulative) — Lower tail of a variance test
- **CHISQ.DIST.RT**(x, deg_freedom) — p-value for chi-square tests with any df
- **CHISQ.INV**(probability, deg_freedom) — Upper limit of a variance confidence interval
- **CHISQ.INV.RT**(probability, deg_freedom) — Chi-square table lookups; variance intervals
- **F.DIST**(x, deg_freedom1, deg_freedom2, cumulative) — Lower tail of a variance-ratio test
- **F.DIST.RT**(x, deg_freedom1, deg_freedom2) — ANOVA and regression p-values
- **F.INV**(probability, deg_freedom1, deg_freedom2) — Lower limit of a variance-ratio interval
- **F.INV.RT**(probability, deg_freedom1, deg_freedom2) — F tables for ANOVA
- **BINOM.DIST**(number_s, trials, probability_s, cumulative) — Defectives in a sample, pass or fail data
- **BINOM.DIST.RANGE**(trials, probability_s, number_s, [number_s2]) — Acceptance-sampling ranges in one step
- **BINOM.INV**(trials, probability_s, alpha) — Acceptance numbers and exact critical counts
- **NEGBINOM.DIST**(number_f, number_s, probability_s, cumulative) — Inspections until a set number of finds
- **HYPGEOM.DIST**(sample_s, number_sample, population_s, number_pop, cumulative) — Sampling small lots without replacement
- **POISSON.DIST**(x, mean, cumulative) — Defects per unit, c- and u-chart probabilities
- **EXPON.DIST**(x, lambda, cumulative) — Time between failures, constant failure rate
- **WEIBULL.DIST**(x, alpha, beta, cumulative) — Life data and wear-out reliability
- **LOGNORM.DIST**(x, mean, standard_dev, cumulative) — Repair times, particle sizes, skewed data
- **LOGNORM.INV**(probability, mean, standard_dev) — Percentile limits for skewed data
- **GAMMA.DIST**(x, alpha, beta, cumulative) — Waiting times for several events
- **GAMMA.INV**(probability, alpha, beta) — Percentiles of gamma-shaped data
- **GAMMA**(number) — Building other distributions by hand
- **GAMMALN**(x) — Large factorials without overflow
- **GAMMALN.PRECISE**(x) — Same use, newer algorithm
- **BETA.DIST**(x, alpha, beta, cumulative, [A], [B]) — PERT task times, proportions
- **BETA.INV**(probability, alpha, beta, [A], [B]) — Exact binomial confidence limits
- **PERMUT**(number, number_chosen) — Counting sequences and orders
- **PERMUTATIONA**(number, number_chosen) — Codes and passwords style counting
- **PROB**(x_range, prob_range, [lower_limit], [upper_limit]) — Custom discrete distributions
- **COMBIN**(number, number_chosen) HELPER — Counting samples where order does not matter
- **COMBINA**(number, number_chosen) HELPER — Counting with replacement

03 STATISTICAL INFERENCE

How sure am I? · 2 stations

- **CONFIDENCE.T**(alpha, standard_dev, size) — Confidence interval for a mean, sigma unknown
- **CONFIDENCE.NORM**(alpha, standard_dev, size) — Confidence interval for a mean, sigma known

04 HYPOTHESIS TESTS

Is the difference real? · 4 stations

- **T.TEST**(array1, array2, tails, type) — Comparing two means
- **Z.TEST**(array, x, [sigma]) — Mean against a target when sigma is known
- **F.TEST**(array1, array2) — Checking variances before choosing T.TEST type
- **CHISQ.TEST**(actual_range, expected_range) — Counts by category; contingency tables

05 CORRELATION

Do they move together? · 7 stations

- **CORREL**(array1, array2) — Strength and direction of a straight-line link
- **PEARSON**(array1, array2) — Legacy sheets; identical result
- ○ **RSQ**(known_y's, known_x's) ⇄ 06 — Share of variation explained by one x
- **COVARIANCE.S**(array1, array2) — Building r by hand on sample data
- **COVARIANCE.P**(array1, array2) — Matches the ToolPak Covariance tool
- **FISHER**(x) — Confidence interval on r; comparing two r values
- **FISHERINV**(y) — Second half of the r interval

06 REGRESSION

Can I model Y from X? · 7 stations

- ○ **LINEST**(known_y's, [known_x's], [const], [stats]) ⇄ 10 — Simple or multiple regression with inference
- **SLOPE**(known_y's, known_x's) — Change in y per unit of x
- **INTERCEPT**(known_y's, known_x's) — y when x is zero
- **STEYX**(known_y's, known_x's) — Width of prediction intervals
- ○ **TREND**(known_y's, [known_x's], [new_x's], [const]) ⇄ 07 — Fitted values, residuals, predictions
- **LOGEST**(known_y's, [known_x's], [const], [stats]) — Growth or decay curves
- **GROWTH**(known_y's, [known_x's], [new_x's], [const]) — Predictions from an exponential model

07 FORECASTING

What comes next? · 6 stations

- **FORECAST.LINEAR**(x, known_y's, known_x's) — Straight-line trend forecasts
- **FORECAST**(x, known_y's, known_x's) — Older workbooks only
- **FORECAST.ETS**(target_date, values, timeline, [seasonality], [data_completion], [aggregation]) — Time series with trend and seasonality
- **FORECAST.ETS.CONFINT**(target_date, values, timeline, [confidence_level], [seasonality], [data_completion], [aggregation]) — Error bands around ETS forecasts
- **FORECAST.ETS.SEASONALITY**(values, timeline, [data_completion], [aggregation]) — Confirming the cycle before trusting a forecast
- **FORECAST.ETS.STAT**(values, timeline, statistic_type, [seasonality], [data_completion], [aggregation]) — Judging forecast accuracy

08 EXPLORATORY DATA ANALYSIS

What shape is it in? · 18 stations

- ○ **FREQUENCY**(data_array, bins_array) ⇄ 02 — Histograms and goodness-of-fit counts
- **QUARTILE.EXC**(array, quart) — Box plots that match Minitab
- **QUARTILE.INC**(array, quart) — Matching older Excel workbooks
- **PERCENTILE.EXC**(array, k) — P5 or P95 limits that match Minitab
- **PERCENTILE.INC**(array, k) — Percentiles that include 0 and 1
- **PERCENTRANK.EXC**(array, x, [significance]) — Where one value sits in the data
- **PERCENTRANK.INC**(array, x, [significance]) — Same, inclusive method
- **RANK.AVG**(number, ref, [order]) — Normal probability plots and rank tests
- **RANK.EQ**(number, ref, [order]) — League tables and Pareto ordering
- ○ **STANDARDIZE**(x, mean, standard_dev) ⇄ 02 — Flagging points beyond three standard deviations
- **GROUPBY**(row_fields, values, function, [field_headers], ...) HELPER — Stratify by heat, shift, line or supplier
- **PIVOTBY**(row_fields, col_fields, values, function, ...) HELPER — Side-by-side comparisons such as LPA vs TWA
- **PERCENTOF**(data_subset, data_all) HELPER — Pareto-style contributions
- **FILTER**(array, include, [if_empty]) HELPER — Slicing a subgroup without copying
- **UNIQUE**(array, [by_col], [exactly_once]) HELPER — Listing categories before grouping
- **SORT**(array, [sort_index], [sort_order], [by_col]) HELPER — Ordering data for plots and ranks
- **SUBTOTAL**(function_num, ref1, [ref2], ...) HELPER — Summaries that follow a filter
- **AGGREGATE**(function_num, options, ref1, [ref2], ...) HELPER — Robust summaries on messy data

09 WHAT-IF & OPTIMISATION

What settings hit the target? · 18 stations

- **Goal Seek** · Data › What-If Analysis › Goal Seek — Which sigma gives Cpk 1.33?
- **Data Table (one variable)** · Data › What-If Analysis › Data Table (column input cell) — Sensitivity of cost or yield to one input
- **Data Table (two variables)** · Data › What-If Analysis › Data Table (row and column input cells) — Cpk map across mean and sigma
- **Scenario Manager** · Data › What-If Analysis › Scenario Manager — Best, likely and worst cases
- **Solver (Simplex LP)** · Data › Solver › Select a Solving Method: Simplex LP — Blends, product mix, scheduling
- **Solver (GRG Nonlinear)** · Data › Solver › Select a Solving Method: GRG Nonlinear — Response optimization, curve fitting
- **Solver (Evolutionary)** · Data › Solver › Select a Solving Method: Evolutionary — Models with IF, LOOKUP or rounding
- **Forecast Sheet** · Data › Forecast › Forecast Sheet — One-click seasonal forecast
- **RAND()** HELPER — Driving simulations through .INV functions
- **RANDARRAY**([rows], [columns], [min], [max], [integer]) HELPER — Thousands of draws in one formula
- **RANDBETWEEN**(bottom, top) HELPER — Random sampling of rows
- **SEQUENCE**(rows, [columns], [start], [step]) HELPER — Input grids for sensitivity tables
- **LET**(name1, name_value1, calculation_or_name2, ...) HELPER — Readable one-cell simulations
- **LAMBDA**([parameter1, ...], calculation) HELPER — Custom test statistics and models
- **MAP**(array1, lambda_or_array) HELPER — Formula-based data tables that recalculate fast
- **SUMPRODUCT**(array1, [array2], ...) HELPER — Objective and constraint cells for Solver
- **MMULT**(array1, array2) HELPER — Regression by matrices; linear models
- **MINVERSE**(array) HELPER — Normal equations for regression by hand

10 ANALYSIS TOOLPAK

Want the full output table? · 19 stations

- **Anova: Single Factor** · Data › Data Analysis › Anova: Single Factor — Three or more group means
- **Anova: Two-Factor With Replication** · Data › Data Analysis › Anova: Two-Factor With Replication — Two factors with repeats
- **Anova: Two-Factor Without Replication** · Data › Data Analysis › Anova: Two-Factor Without Replication — Randomized block designs
- **t-Test: Paired Two Sample for Means** · Data › Data Analysis › t-Test: Paired Two Sample for Means — Before and after on the same units
- **t-Test: Two-Sample Assuming Equal Variances** · Data › Data Analysis › t-Test: Two-Sample Assuming Equal Variances — Independent groups, equal spread
- **t-Test: Two-Sample Assuming Unequal Variances** · Data › Data Analysis › t-Test: Two-Sample Assuming Unequal Variances — Independent groups, unequal spread
- **z-Test: Two Sample for Means** · Data › Data Analysis › z-Test: Two Sample for Means — Two means with known variances
- **F-Test Two-Sample for Variances** · Data › Data Analysis › F-Test Two-Sample for Variances — Comparing two variances
- **Descriptive Statistics** · Data › Data Analysis › Descriptive Statistics — One-click summary
- **Regression** · Data › Data Analysis › Regression — Full regression output
- **Correlation** · Data › Data Analysis › Correlation — Several variables at once
- **Covariance** · Data › Data Analysis › Covariance — Several variables at once
- **Histogram** · Data › Data Analysis › Histogram — Quick frequency table
- **Rank and Percentile** · Data › Data Analysis › Rank and Percentile — Relative standing table
- **Random Number Generation** · Data › Data Analysis › Random Number Generation — Static simulated data
- **Sampling** · Data › Data Analysis › Sampling — Drawing an audit sample
- **Moving Average** · Data › Data Analysis › Moving Average — Trend smoothing
- **Exponential Smoothing** · Data › Data Analysis › Exponential Smoothing — Level-only forecasts
- **Fourier Analysis** · Data › Data Analysis › Fourier Analysis — Vibration and periodic signals

11 LEGACY NAMES (CLOSED LINE)

Still runs; use the new name · 39 stations

- ~~BETADIST~~ → BETA.DIST
- ~~BETAINV~~ → BETA.INV
- ~~BINOMDIST~~ → BINOM.DIST
- ~~CHIDIST~~ → CHISQ.DIST.RT
- ~~CHIINV~~ → CHISQ.INV.RT
- ~~CHITEST~~ → CHISQ.TEST
- ~~CONFIDENCE~~ → CONFIDENCE.NORM
- ~~COVAR~~ → COVARIANCE.P
- ~~CRITBINOM~~ → BINOM.INV
- ~~EXPONDIST~~ → EXPON.DIST
- ~~FDIST~~ → F.DIST.RT
- ~~FINV~~ → F.INV.RT
- ~~FORECAST~~ → FORECAST.LINEAR
- ~~FTEST~~ → F.TEST
- ~~GAMMADIST~~ → GAMMA.DIST
- ~~GAMMAINV~~ → GAMMA.INV
- ~~HYPGEOMDIST~~ → HYPGEOM.DIST
- ~~LOGINV~~ → LOGNORM.INV
- ~~LOGNORMDIST~~ → LOGNORM.DIST
- ~~MODE~~ → MODE.SNGL
- ~~NEGBINOMDIST~~ → NEGBINOM.DIST
- ~~NORMDIST~~ → NORM.DIST
- ~~NORMINV~~ → NORM.INV
- ~~NORMSDIST~~ → NORM.S.DIST
- ~~NORMSINV~~ → NORM.S.INV
- ~~PERCENTILE~~ → PERCENTILE.INC
- ~~PERCENTRANK~~ → PERCENTRANK.INC
- ~~POISSON~~ → POISSON.DIST
- ~~QUARTILE~~ → QUARTILE.INC
- ~~RANK~~ → RANK.EQ
- ~~STDEV~~ → STDEV.S
- ~~STDEVP~~ → STDEV.P
- ~~TDIST~~ → T.DIST.RT or T.DIST.2T
- ~~TINV~~ → T.INV.2T
- ~~TTEST~~ → T.TEST
- ~~VAR~~ → VAR.S
- ~~VARP~~ → VAR.P
- ~~WEIBULL~~ → WEIBULL.DIST
- ~~ZTEST~~ → Z.TEST

12 BEYOND EXCEL (WALK FROM HERE)

Use Minitab or Python · 12 stations

- **Normality test (Anderson-Darling, Shapiro-Wilk)** — Workaround: Normal probability plot with NORM.S.INV and RANK.AVG; check SKEW and KURT
- **One- and two-proportion tests** — Workaround: z statistic with NORM.S.DIST; exact test with BINOM.DIST
- **Nonparametric tests (sign, Wilcoxon, Mann-Whitney, Kruskal-Wallis)** — Workaround: Sign test with BINOM.DIST; rank sums with RANK.AVG
- **Equal variances, three or more groups (Levene, Bartlett)** — Workaround: One-way ANOVA on absolute deviations from each group median
- **Post-hoc comparisons (Tukey)** — Workaround: Fisher LSD with T.INV.2T; no studentized range in Excel
- **Unbalanced or multi-factor ANOVA and DOE analysis** — Workaround: LINEST with coded columns for two-level factorials
- **Power and sample size** — Workaround: Sample-size formula with NORM.S.INV
- **Process capability** — Workaround: Cpk from AVERAGE and STDEV.S, or R-bar over d2
- **Control charts** — Workaround: Limits from AVERAGE and constants, drawn as a line chart
- **Gage R&R** — Workaround: ANOVA method built from DEVSQ
- **Weibull fitting** — Workaround: LINEST on the linearized Weibull plot
- **Logistic or Poisson regression** — Workaround: Solver maximizing the log-likelihood
- **PY**(python_code, return_type) HELPER — scipy.stats and statsmodels for tests Excel lacks

Mind the gap: Z.TEST returns the upper one-tail p · CONFIDENCE.T is the ± half-width, not the interval · STDEV.S is overall σ (Ppk, not Cpk)

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